

SQL Subquery Practice Kit - 20 Questions + Dataset

BEGINNER

1. Find employees earning above the average salary.
2. Find employees working in the same department as Ravi.
3. Find the highest-paid employee.
4. Find employees hired after Amit.
5. Find employees earning less than the highest salary.

INTERMEDIATE

6. Find employees who are managers.
7. Find employees without any projects.
8. Find employees whose salary is above their department average.
9. Find departments with more than two employees.
10. Find projects handled by employees earning above the company average.
11. Find employees who work in departments located in London.
12. Find employees earning more than at least one employee in Finance.
13. Find employees hired before everyone in the Sales department.

ADVANCED

14. Find employees earning more than their manager.
15. Find the employee(s) with the highest salary in each department.
16. Find departments where no employee earns less than \$60,000.
17. Find employees who belong to the department with the highest payroll.
18. Find employees who have projects with a budget greater than the average project budget.
19. Find employees whose salary is higher than the average salary of all managers.
20. Find the employee(s) with the third-highest salary in the company.

DATABASE DATASET

Departments Table

```
CREATE TABLE departments (  
    dept_id INT PRIMARY KEY,  
    dept_name VARCHAR(50),  
    location VARCHAR(50)  
);
```

```
INSERT INTO departments VALUES  
(1,'HR','New York'),  
(2,'IT','London'),  
(3,'Finance','Chicago'),  
(4,'Sales','Boston');
```

Employees Table

```
CREATE TABLE employees (  
    emp_id INT PRIMARY KEY,  
    emp_name VARCHAR(50),  
    dept_id INT,  
    manager_id INT,  
    salary INT,  
    hire_date DATE  
);  
  
INSERT INTO employees VALUES  
(101,'Amit',1,NULL,50000,'2020-01-15'),  
(102,'Ravi',2,101,70000,'2019-05-20'),  
(103,'Neha',2,101,65000,'2021-02-10'),  
(104,'Priya',3,102,80000,'2018-07-01'),  
(105,'Arjun',3,102,90000,'2017-09-15'),  
(106,'Karan',4,103,60000,'2022-03-18'),  
(107,'Meera',4,103,62000,'2021-11-12'),  
(108,'Sneha',2,101,72000,'2019-12-01'),  
(109,'Rahul',1,NULL,55000,'2023-04-08'),  
(110,'Vikas',3,104,85000,'2020-08-22');
```

Projects Table

```
CREATE TABLE projects (  
    project_id INT PRIMARY KEY,  
    project_name VARCHAR(50),  
    emp_id INT,  
    budget INT  
);  
  
INSERT INTO projects VALUES  
(1,'ERP',102,500000),  
(2,'CRM',103,300000),  
(3,'AI Platform',104,900000),  
(4,'Analytics',105,700000),  
(5,'HR Portal',107,200000),  
(6,'Sales App',110,600000);
```